



MPAIRWE LAUBEN

DEVSECOPS · CLOUD-NATIVE · ML ·
AGENTIC AI ENGINEER

ABOUT

DevSecOps engineer and applied ML practitioner. Co-founder of IONATECH (ionatec.com), an AI startup, and engineering partner at Sauti Health (sautihealth.com), building emergency-dispatch infrastructure for sub-Saharan Africa. By day, CI/CD, cloud, and security posture at AIBOS Uganda. Outside: training models, building agentic systems on LangChain and LangGraph, and instrumenting MCP servers that expose tools to LLMs.

📍 Kampala, Uganda 📞 +256 773 336 896 ✉️ mpairwelauben75@gmail.com

WORK EXPERIENCE

AIBOS Uganda

Kampala

Feb 2024 – Present

DEVSECOPS ENGINEER

- ▶ Own CI/CD posture on GitHub Actions, cutting deploy time ~60% while adding SBOM generation, container scanning, and signature verification
- ▶ Run infrastructure-as-code (Terraform) across AWS, Azure, and GCP with policy-as-code gates on every plan
- ▶ Harden Kubernetes workloads with Pod Security Standards, NetworkPolicies, and supply-chain controls (cosign / Trivy)
- ▶ Wire Prometheus + Grafana + Loki dashboards with SLO-driven alerts and on-call runbooks
- ▶ Drive compliance posture (encryption, RBAC, secrets via Vault) across the cloud estate

GITHUB ACTIONS · TERRAFORM ·
KUBERNETES · TRIVY · COSIGN · VAULT ·
PROMETHEUS · GRAFANA · AWS

SKILLS

TRIVY GRYPE SNYK COSIGN SIGSTORE
SLSA OPA / GATEKEEPER KYVERNO
FALCO VAULT EXTERNAL SECRETS
POD SECURITY STANDARDS
NETWORKPOLICIES GITHUB ACTIONS
ARGOCD FLUX TERRAFORM PULUMI
MLFLOW WEIGHTS & BIASES DVC FEAST
KUBEFLOW RAY TRITON BENTOML
EVIDENTLY SCIKIT-LEARN TENSORFLOW
PYTORCH PANDAS NUMPY LANGCHAIN
LANGGRAPH MODEL CONTEXT PROTOCOL
VERCEL AI GATEWAY AI SDK V6 QDRANT
PGVECTOR BRAINTRUST PROMPTFOO
LANGSMITH DEEPEVAL GUARDRAILS AWS
GCP AZURE VERCEL KUBERNETES DOCKER
HELM PROMETHEUS GRAFANA LOKI
OPENTELEMETRY ISTIO POSTGRESQL
PGVECTOR MYSQL MONGODB REDIS
QDRANT SNOWFLAKE FIREBASE SUPABASE
NEXT.JS REACT TYPESCRIPT TAILWIND
SHADCN/UI MOTION NODE.JS FASTAPI
SPRING BOOT TRPC FLUTTER
REACT NATIVE KOTLIN SWIFT FIREBASE
APP STORE / PLAY

LINKS

-  LINKEDIN
mpairwe-lauben-3a1461344
-  GITHUB
github.com/mpairweLandwind
-  IONATECH
ionatec.com
-  SAUTI HEALTH
sautihealth.com

LANGUAGES

English (Fluent)
German (Basic)

Sauti Health

Kampala

2026 – Present

IONATECH

Kampala

Feb 2023 – Present

DEVELOPER · ENGINEERING PARTNER

- ▶ Partner with the Sauti Health team on the engineering behind a pre-hospital emergency dispatch platform for sub-Saharan Africa
- ▶ Build and harden backend services that coordinate community responders and motorcycle riders over USSD on basic 2G phones
- ▶ Ship offline-first responder workflows on low-end Android, with sync, retries, and conflict resolution for patchy networks
- ▶ Integrate mobile-money payment flows (MTN MoMo · Airtel Money) and DHIS2 reporting for health-system interoperability
- ▶ Instrument telemetry, dashboards, and CI gates on safety-critical paths so dispatch reliability stays measurable

PYTHON · FASTAPI · POSTGRESQL · USSD/IVR · MOBILE MONEY · DHIS2 · OFFLINE-FIRST · OBSERVABILITY

CO-FOUNDER · ML & AGENTIC ENGINEER

- ▶ Co-founded IONATECH (ionatec.com) — a startup building applied ML, agentic AI, and secure cloud systems for East African operators
- ▶ Ship LLM agents with LangChain + LangGraph: tool-use, retrieval, multi-agent orchestration, and evals
- ▶ Build RAG pipelines on Qdrant / pgvector with hybrid retrieval, rerank, and Braintrust/ Promptfoo eval harnesses
- ▶ Deliver classical ML for classification, regression, and ranking — with monitoring, drift detection, and retraining loops
- ▶ Run the engineering curriculum and mentor residents from prototype through production

LANGCHAIN · LANGGRAPH · MCP · QDRANT · BRAINTRUST · PYTHON · FASTAPI · NEXT.JS · SCIKIT-LEARN

PERSONAL DETAILS

DATE OF BIRTH
22 October 1995

NATIONALITY
Ugandan

LOCATION
Kampala, Uganda

REFERENCES

Dr. Kimbugwe Nasser
Lecturer & Academic Supervisor
Makerere University, School of Computing & IT

Kalema Martin
Supervisor
AIBOS Uganda

Dr. Mark Magumba
Senior Lecturer
Makerere University, School of Computing & IT

Fahadi Nguyenguye
Technical Lead
PAHAPPA LIMITED

Dr. Ggaliwango Marvin
Lecturer & AI/ML Mentor
Makerere University, School of Computing & IT

Makerere University
Kampala
Jan 2022 – Present

PAHAPPA LIMITED
Kampala
Jun 2022 – Jan 2024

WEB INFO-NET LIMITED
Kampala
Jun 2018 – Aug 2022

STUDENT SOFTWARE ENGINEER

- ▶ Built TaskApp — a Kotlin task management app with real-time Firebase sync
- ▶ Developed scalable cross-platform apps using React Native and Flutter
- ▶ Implemented secure JWT and biometric authentication flows
- ▶ Delivered a fintech mobile app with digital payments, rewards, and transaction tracking
- ▶ Integrated third-party payment APIs and designed scalable backend architecture

KOTLIN · REACT NATIVE · FLUTTER · FIREBASE · JWT · FINTECH

DEVOPS INTERN → ENGINEER

- ▶ Built enterprise applications with Spring Boot (MVC, Data JPA, Security, Cloud)
- ▶ Used Hibernate ORM/JPA with MySQL, PostgreSQL, Oracle, and MongoDB databases
- ▶ Designed Microservice and Monolithic architectures based on project requirements
- ▶ Developed RESTful APIs and event-driven systems for high-throughput workflows

SPRING BOOT · JAVA · HIBERNATE · MYSQL · POSTGRESQL · MICROSERVICES · REST API

APPLICATION DEVELOPER INTERN & TRAINER

- ▶ Developed mobile apps using Swift, Java, and Kotlin for iOS and Android platforms
- ▶ Built an Odoo-based Employment Management System for HR automation
- ▶ Implemented secure authentication and third-party API integrations
- ▶ Provided Level 1 technical support, troubleshooting, and end-user training

SWIFT · JAVA · KOTLIN · OD00 · IOS · ANDROID

ZALONIDENTALHUB

Kampala

Apr 2022 – Present

LEADING DEVELOPER & MAINTAINER

- ▶ Led full SDLC for dental practice management software from conception to delivery
- ▶ Built C#/VB.NET desktop systems for inventory and patient management
- ▶ Developed a Flutter mobile app with e-commerce, appointments, and booking features
- ▶ Ensured seamless desktop-mobile data interoperability
- ▶ Led a team of developers, defining software architectures and coding standards

C# · VB.NET · FLUTTER · DESKTOP · MOBILE · TEAM LEAD

EDUCATION

Makerere University

2022 – Expected
2027

CGPA 4.31

Bachelor of Science in Software Engineering

- ▶ Specialising in AI/ML and Enterprise Application Development
- ▶ Integrated Supabase, MongoDB, Cloudflare, and Firebase for scalable storage
- ▶ Implemented RBAC and encryption for secure multi-role systems
- ▶ Built a full HRMS system with advanced role and permissions management
- ▶ Worked on IoT and Embedded Systems research projects
- ▶ Optimised PostgreSQL queries for high-performance data retrieval

Makerere University

2019 – 2023

Bachelor of Science in Computer Engineering

- ▶ Transitioned to the Software Engineering programme
- ▶ Studied Digital Electronics and hardware design principles
- ▶ Hardware repair, maintenance, and systems diagnostics

**Uganda Technical
College – Lira**

2016 – 2018

CGPA 4.77

**National Diploma in Information
Communication Technology**

- ▶ Database Management and Web Development
- ▶ Internet technologies and networking fundamentals
- ▶ RBAC and Encryption implementation
- ▶ Java programming and object-oriented design
- ▶ Networking and Server Configuration

**Boston High School –
Mpala**

2011 – 2015

**Uganda Certificate of Education
(UCE)**

- ▶ Graduated in Top 5% of class
- ▶ Varsity Debate Team Captain

CERTIFICATIONS

PLANNED	AWS Certified DevOps Engineer · Professional	— Amazon Web Services
PLANNED	Certified Kubernetes Security Specialist	— CNCF
PLANNED	Certified Kubernetes Administrator	— CNCF
PLANNED	HashiCorp Certified · Terraform Associate	— HashiCorp
PLANNED	AWS Solutions Architect · Associate	— Amazon Web Services
PLANNED	Google Cloud Professional ML Engineer	— Google Cloud